

Prerana Sodankoor

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Education

PES University

B.Tech, Computer Science & Engineering / CGPA: **8.73** (Sem 6 SGPA: **9.08**)

Bengaluru, India

2023 – Present

Whitefield Global School

Pre-University (PCMB) – **95%**

Class XII

Experience

Research Intern

Center for Computer Networks & Cybersecurity, PES University

Bengaluru, India

Jun 2025 – Jul 2025

- Researched and implemented a security pipeline for LLMs using ML models trained for threat classification and Data Loss Prevention (DLP) to prevent confidential data leakage.
- Authored and presented a research paper at **IEEE**; accepted for publication. Received IEEE certification.
- Reviewed 2 papers for the CMT conference.

Projects

Secure Prompt

Python, DistilBERT, LangChain, RAG, Streamlit

- Security pipeline to detect and mitigate prompt injection attacks in LLMs using static semantic analysis and a fine-tuned DistilBERT classifier integrated into a RAG pipeline with a Streamlit frontend.

AI Chatbot

Python, LLaMA, LangChain

- Conversational assistant using LLaMA and LangChain capable of maintaining context across multi-turn conversations and generating coherent real-time responses.

Resume Critique Tool

Python, NLP

- AI-powered system that compares resumes against job descriptions and delivers targeted, actionable feedback using NLP to help candidates tailor applications.

Mini Docksmith

C, Linux

- Lightweight Docker replica with an interactive dashboard supporting core containerisation commands; hands-on environment for learning container concepts.

SchedViz

TypeScript, HTML/CSS

- Interactive visualisation platform for OS process scheduling algorithms with step-by-step animated walkthroughs.

Stock & Inventory Management System

Python, SQL

- DBMS application with full CRUD operations for managing stock and inventory, augmented with voice-to-text support for natural language querying.

Technical Skills

Languages: Python, C, JavaScript, SQL

Security: Prompt Injection, DLP, RAG Security, Wireshark, Ethical Hacking

AI/ML: NLP, Transformers, LLMs, LangChain, DistilBERT

Systems & Tools: Linux, Git/GitHub, Docker

Web: HTML, CSS, MERN Stack

Achievements & Leadership

- **2nd Place – Codeventure 3.0 CTF** (Capture The Flag), ACM-W Club, PES University – September 2025
- **IEEE Published Researcher** – Paper on securing LLMs using ML & DLP; presented and certified at IEEE
- **Merit & CNR Scholarship** – Recognized for consistent academic excellence at PES University
- **Design Head & Core Member**, AiCoding Club; **Member**, ACM-W Club, PES University
- Participated in **IEEE L8 CTF**, Heal-O Code Hackathon, and Delta-Time Game Dev Hackathon
- Team Lead for multiple academic group projects

Certifications

Introduction to Cybersecurity | Introduction to AI | ML using Python | Trust & Security in Google Cloud | Introduction to Ethical Hacking | Cybersecurity Analyst Job Simulation – Tata (Forge)